

Probability

ANSWER KEY



Basic probability

- 1 Two fair six-sided dice are rolled. Find the probability their sum is 8.

Total outcomes: $n(S) = 36$. Favorable outcomes (sum = 8): (2, 6), (3, 5), (4, 4), (5, 3), (6, 2) — that's 5.

$$\text{So } P(\text{sum} = 8) = \frac{5}{36}.$$

Complement rule

- 2 If $P(E) = 0.35$, find $P(E')$.

By Definition 6.2, $P(E') = 1 - P(E) = 1 - 0.35 = 0.65$.

Permutations and combinations

- 3 How many ways can a committee of 4 be chosen from 10 people?

Order doesn't matter, so use combinations: $\binom{10}{4} = \frac{10!}{4!6!} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = 210$.

- 4 How many ways can 2 officers (president, vice president — order matters) be chosen from 6 candidates?

Order matters, so use permutations: $P(6, 2) = \frac{6!}{4!} = 6 \cdot 5 = 30$.

- 5 Explain, using Theorem 6.1, how $\binom{6}{2}$ relates to $P(6, 2)$ from the previous question.

Theorem 6.1 says $\binom{6}{2} = \frac{P(6, 2)}{2!} = \frac{30}{2} = 15$ — combinations undercount permutations by exactly the 2! orderings of each chosen pair.

Independent events and the addition rule

- 6 Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

By Definition 6.4, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.4)(0.5) = 0.2$.

- 7 A card is drawn from a standard 52-card deck. Find $P(\text{Queen or Spade})$.

$P(\text{Queen}) = \frac{4}{52}$, $P(\text{Spade}) = \frac{13}{52}$, and $P(\text{Queen and Spade}) = \frac{1}{52}$ (the Queen of Spades).

By the addition rule: $P(\text{Queen or Spade}) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$.